

About the Prediction Probability & Uncertainty Standard

What Is the Prediction Probability & Uncertainty Standard?

The Prediction Probability & Uncertainty Standard (PPUS) is the PGA™ Parent Standard governing how probability, uncertainty, confidence, ranges, conditionality, and related predictive qualification are represented around a Governed Prediction.

Its purpose is not to determine the future.

Its purpose is to ensure that the strength of a prediction is represented according to what the underlying methodology and predictive conditions legitimately support.

A prediction may state:

70% probability

high confidence

likely

10–15% expected range

low uncertainty

These expressions are not interchangeable.

PPUS establishes the governance structure needed to preserve their different meanings.

Probability ≠ Confidence ≠ Certainty

Why the Standard Exists

Predictive systems increasingly produce precise-looking numbers.

A percentage, probability, confidence score, or narrow range can create an impression of certainty even where substantial uncertainty remains.

This becomes particularly important in AI systems, forecasting platforms, financial models, prediction markets, risk systems, and automated decision environments.

A system may display:

82%

without making clear whether that number represents:

event probability; model confidence; classification confidence; market-implied probability; expert judgment; calibrated probability; or another predictive quantity.

The number alone does not establish its meaning.

Same Probability Value ≠ Same Predictive Meaning

PPUS exists to prevent predictive strength from being created by notation rather than methodology.

What PPUS Governs

PPUS governs the representation and predictive meaning of:



probability; uncertainty; confidence; probability ranges; predictive ranges; intervals; distributions; conditional probability; scenario-dependent probability; qualitative likelihood; probability provenance; probability transformation; precision and rounding;

uncertainty qualification; historical probability and uncertainty states.

It ensures that these representations remain connected to the prediction they qualify.

Probability Is Not Certainty

A prediction with:

95% probability

does not establish that the outcome will occur.

Likewise:

5% probability

does not establish that the outcome cannot occur.

Probability represents estimated likelihood under governed predictive conditions.

It does not transform a future event into a known fact.

High Probability $\neq$ Certainty

Probability Is Not Confidence

Probability and confidence describe different dimensions of a prediction.

A system may estimate:

80% probability

while having:

low confidence in that estimate

because the Prediction Basis is limited, assumptions are unstable, or the methodology is being applied outside its strongest operating conditions.

Conversely, a system may have high confidence that an event has only a 5% probability.

PPUS preserves these dimensions separately.

High Probability $\neq$ High Confidence

Uncertainty Does Not Disappear When a Number Appears

Quantification can make uncertainty easier to represent.

It does not eliminate uncertainty.

Predictive uncertainty may arise from:

incomplete information; unstable assumptions; model limitations; environmental change; parameter uncertainty; structural uncertainty; behavioral uncertainty; scenario uncertainty; temporal uncertainty; unknown dependencies.

Some uncertainty can be quantified.

Some can only be bounded.

Some may need to remain explicitly unquantified.

Unquantified Uncertainty $\neq$ Nonexistent Uncertainty

Probability Requires Context

A probability becomes meaningful only in relation to the prediction it qualifies.

Its interpretation may depend upon:

Predictive Claim; methodology; Prediction Horizon; Predictive Boundary; assumptions; conditions; scenario; probability source.

For example:

70%

for an event occurring within seven days is not the same predictive representation as:

70%

for that event occurring within five years.

Probability at Horizon A $\neq$ Probability at Horizon B

Probability Provenance Matters

Two systems may produce exactly the same probability value through completely different mechanisms.

One may derive it from:

statistical modeling;

another from:

expert judgment;

another from:

simulation;

another from:

a prediction market;

another from:

an AI model score transformed into probability.

PPUS requires the origin and meaning of probability to remain distinguishable.

Same Probability Value $\neq$ Same Predictive Meaning

Scores Are Not Automatically Probabilities

AI systems frequently produce scores between 0 and 1.

That numerical format may look like probability.

It does not automatically mean the score represents the probability of a real-world future event.

A classifier confidence score, anomaly score, similarity score, ranking score, or internal model value may require additional methodology before it can legitimately be represented as predictive probability.

Score $\neq$ Probability by Default

Model Confidence $\neq$ Event Probability

Ranges Require Meaning

A range may represent:

expected outcome variation; predictive interval; confidence interval; scenario range; probability range; tolerance range.

These concepts are not equivalent.

PPUS requires the meaning of the range to remain identifiable.

Confidence Interval $\neq$ Predictive Interval

A narrower range also does not automatically indicate a better prediction.

A range can become artificially narrow simply because uncertainty was omitted.

Narrower Range $\neq$ Better Prediction by Default

Conditional Probability Must Preserve Its Conditions

A probability may depend upon specific assumptions or conditions.

For example:

80% probability of project completion by December if financing remains available.

Removing the financing condition changes the predictive meaning.

It must not silently become:

80% probability of project completion by December.

Conditional Probability Without Its Condition $\neq$ Same Probability

Unknown Probability Is a Legitimate Result

Not every prediction supports defensible numerical probability.

Where methodology cannot establish a meaningful probability, the legitimate result may be:

Probability Unable to Determine

rather than an invented percentage.

In particular:

Unknown Probability $\neq$ 50% Probability

A 50% probability is a predictive estimate.

It is not a universal representation of ignorance.

Probability Changes Over Time

Probability may be legitimate when initially established and later become outdated as:

Prediction Basis changes; assumptions change; operating conditions change; new information becomes available; the Prediction Horizon approaches; the environment shifts.

Therefore:

Valid Initial Probability $\neq$ Valid Continuing Probability

PPUS preserves the probability and uncertainty state associated with the prediction at the relevant point in its lifecycle.

Historical states must not be silently replaced by later estimates.

What PPUS Does Not Govern

PPUS does not:

create the Predictive Claim; determine Prediction Methodology; establish Prediction Basis; identify Predictive Assumptions; determine whether an outcome is feasible; aggregate multiple predictors into a Collective Prediction; determine how a prediction may ultimately be relied upon; determine Operational Reality; determine whether the prediction was correct; determine predictive performance.

These spaces remain governed elsewhere within PGA™ and the wider OOF® architecture ecosystem.

Position Within PGA™

PPUS operates after a specific Governed Prediction has been formed.

The complete PGA™ Governance Flow is:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → [Prediction Probability & Uncertainty] → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

PPUS therefore answers a different question from Prediction Formation.

PFOS establishes:

What is the prediction?

PPUS establishes:

How strongly may that prediction legitimately be expressed?

Why PPUS Matters

Predictive systems increasingly influence:

investments; healthcare; industrial operations; infrastructure; supply chains; autonomous systems; strategic planning; financial decisions; public policy.

In these environments, the difference between:

80% probability

and:

80% model confidence

can materially change a decision.

PPUS provides a common governance structure through which humans, AI systems, organizations, and automated processes can distinguish those meanings before relying upon them.

Use Case 1 — AI Displays “90% Confidence”

A healthcare prediction system states:

90% confidence — patient deterioration expected within 48 hours.

Clinicians interpret this as:

90% probability that deterioration will occur.

The technical team later establishes that 90% represents an internal classification-confidence score rather than a calibrated probability of the real-world event.

The critical governance question is:

What does the 90% actually represent?

PPUS requires the system to distinguish the Predictive Claim, event probability, model confidence, uncertainty, methodology, and applicable conditions.

The score may remain useful.

Its meaning cannot silently change because users interpret it as probability.

Model Confidence $\neq$ Event Probability

Use Case 2 — Two Predictions Both Say 70%

An investment committee receives two predictions concerning whether a major project will become operational before the end of 2028.

Prediction A states:

70%

based on a documented statistical methodology and historical project data.

Prediction B also states:

70%

but represents the average subjective estimate of several executives.

The identical numbers create an appearance of independent confirmation.

The critical governance question is:

Do identical probability values represent equivalent predictive information when their origins and methodologies are materially different?

PPUS requires both predictions to preserve their probability provenance, methodology, uncertainty, confidence meaning, assumptions, and conditions.

They may legitimately be compared.

They must not become methodologically equivalent merely because their displayed numbers are identical.

Same Probability Value $\neq$ Same Predictive Meaning

Canonical Boundary

PPUS governs how probability, uncertainty, confidence, ranges, conditionality, and related predictive qualification are represented in association with a Governed Prediction. It does not determine the Predictive Claim itself, establish Operational Reality, determine Prediction Feasibility, create Collective Prediction, determine reliance, or decide whether the predicted outcome ultimately occurs.

PPUS governs:

how strongly a prediction may legitimately be expressed.

It does not govern:

whether the future will conform to that expression.

Canonical Closing Statement

Probability gives uncertainty a form. It does not make uncertainty disappear.

PPUS ensures that probability, confidence, ranges, and uncertainty preserve their actual predictive meaning rather than becoming interchangeable signals of certainty.

A quantified prediction must not claim more certainty than its methodology can legitimately support.

Governed Links

[About Prediction Probability & Uncertainty Standard \(PPUS\).](#)

[Prediction Probability Representation Module \(PRPM\).](#)

[Prediction Uncertainty Module \(PUNM\).](#)

[Confidence Representation Module \(CFRM\).](#)

[Prediction Range Governance Module \(PRGM\).](#)

[Conditional Probability Module \(CPM\).](#)

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